

Course Code and Title: MAT110 Quantitative Methods**Lesson 2: Week 3****Topic: Central Location and Dispersion****Introduction**

A measure of central tendency is an important aspect of quantitative data. It is an estimate of a “typical” value. A measure of central tendency (also referred to as measures of central location) is a summary measure that attempts to describe a whole set of data with a single value that represents the middle or center of its distribution. They are also classed as summary statistics. The mean (often called the average) is most likely the measure of central tendency that you are most familiar with, but there are others, such as the median and the mode.

The mean, median and mode are all valid measures of central tendency, but under different conditions, some measures of central tendency become more appropriate to use than others. In the following sections, we will look at the mean, mode and median, and learn how to calculate them and under what conditions they are most appropriate to be used.

Measures of Central Location

Mean, median, and mode are three kinds of "averages". There are many "averages" in statistics, but these are the three most common, and are certainly the three you are most likely to encounter.

❖ Mean

An average that uses the exact value of each entry is the mean (sometimes called the arithmetic mean). To compute the mean, we add the values of all the entries and then divide by the number of entries.

$$\text{Mean} = \frac{\text{Sum of all entries}}{\text{Number of entries}}$$

The mean is the average usually used to compute a test average.

How to find the mean?

1. Compute; that is, find the sum of all the data values.
2. Divide the total by the number of data values.

Sample statistic

Population parameter μ

Where n = number of data values in the sample

N = number of data values in the population

Example:

To graduate, Linda needs at least a B in Biology. She did not do very well on her first three tests; however, she did well on the last four. Here are her scores:

58 67 60 84 93 98 100

Compute the mean and determine if Linda's grade will be a B (80 to 89 average) or a C (70 to 79 average).

Solution:

Mean = $\frac{\text{Sum of scores}}{\text{Number of scores}}$

Number of scores

$$\text{Mean} = \frac{58 + 67 + 60 + 84 + 93 + 98 + 100}{7}$$

Mean = 80, since the average is 80, Linda will get the needed B.

❖ Median

Another average that is useful is the median, or central value, of an ordered distribution. When you are given the median, you know there are an equal number of data values in the ordered distribution that are above it and below it. The most important step in finding the median is to first order the data from smallest to largest.

How to find the median?

The median is the central value of an ordered distribution. To find it,

1. Order the data from smallest to largest.
2. For an odd number of data values in the distribution,

Median = Middle data value

3. For an even number of data values in the distribution,

Median = $\frac{\text{Sum of middle two values}}{2}$

The median uses the position rather than the specific value of each data entry. If the extreme values of a data set change, the median usually does not change. This is why the median is often used as the average for house prices. If one mansion costing several million dollars sells in a community of much-lower-priced homes, the median selling price for houses in the community would be affected very little, if at all.

Example:

What do barbecue-flavored potato chips cost? According to Consumer Reports, Volume 66, No. 5, the prices per ounce in cents of the rated chips are

19 19 27 28 18 35

- a. To find the median, we first order the data, and then note that there are an even number of entries. So, the median is constructed using the two middle values.

$$\begin{array}{cccccc} 18 & 19 & 19 & 27 & 28 & 35 \\ & & \diagdown & / & & \\ & & \text{middle values} & & & \end{array}$$
$$\text{Median} = \frac{19 + 27}{2} = 23 \text{ cents}$$

- b. According to Consumer Reports, the brand with the lowest overall taste rating costs 35 cents per ounce. Eliminate that brand, and find the median price per ounce for the remaining barbecue-flavored chips. Again, order the data. Note that there are an odd number of entries, so the median is simply the middle value.

$$\begin{array}{ccccc} 18 & 19 & 19 & 27 & 28 \\ & & \uparrow & & \\ & & \text{middle value} & & \end{array}$$
$$\text{Median} = \text{middle value} = 19 \text{ cents}$$

- c. One ounce of potato chips is considered a small serving. Is it reasonable to budget about \$10.45 to serve the barbecue-flavored chips to 55 people?

Answer: Yes, since the median price of the chips is 19 cents per small serving. This budget for chips

assumes that there is plenty of other food!

❖ **Mode**

The mode is the value that occurs most often in the data. It is important to note that there may be more than one mode in the dataset. Not every data set has a mode. For example, if Professor Fair gives equal numbers of A's, B's, C's, D's, and F's, then there is no modal grade. In addition, the mode is not very stable. Changing just one number in a data set can change the mode dramatically. However, the mode is a useful average when we want to know the most frequently occurring data value, such as the most frequently requested shoe size.

Example:

Find the mode for the following list of values:

13, 18, 13, 14, 13, 16, 14, 21, 13

The mode is the number that is repeated more often than any other, so 13 is the mode.

Resistant Measure

A resistant measure is one that is not influenced by extremely high or low data values. The mean is not a resistant measure of center because we can make the mean as large as we want by changing the size of only one data value. The median, on the other hand, is more resistant. However, a disadvantage of the median is that it is not sensitive to the specific size of a data value.

Trimmed Mean

A measure of center that is more resistant than the mean but still sensitive to specific data values is the trimmed mean. A trimmed mean is the mean of the data values left after trimming a specified percentage of the smallest and largest data values from the data set. Usually, a 5% trimmed mean is used. This implies that we trim the lowest 5% of the data as well as the highest 5% of the data. A similar procedure is used for a 10% trimmed mean.

How to compute a 5% trimmed mean?

1. Order the data from smallest to largest.
2. Delete the bottom 5% of the data and the top 5% of the data. Note: If the calculation of 5% of the number of data values does not produce a whole number, round to the nearest integer.
3. Compute the mean of the remaining 90% of the data.

Example:

Barron's Profiles of American Colleges, 19th Edition, lists average class size for introductory lecture courses at each of the profiled institutions. A sample of 20 colleges and universities in California showed class sizes for introductory lecture courses to be

14	20	20	20	20	23	25	30	30	30
35	35	35	40	40	42	50	50	80	80

(a) Compute the mean for the entire sample.



Add all the values and divide by 20:

$$\bar{x} = \frac{\sum x}{n} = \frac{719}{20} \approx 36.0$$

(b) Compute a 5% trimmed mean for the sample.



The data are already ordered. Since 5% of 20 is 1, we eliminate one data value from the bottom of the list and one from the top. These values are circled in the data set. Then take the mean of the remaining 18 entries.

$$5\% \text{ trimmed mean} = \frac{\sum x}{n} = \frac{625}{18} \approx 34.7$$

Summary

A measure of central tendency or also known as a measure of central location is a summary measure of a dataset through a single value that reflects the center of the data distribution. Commonly used measures of center are the arithmetic mean, the median, and the mode. The arithmetic mean is the average and can be computed by getting the sum of all the entries and then dividing by the number of entries. The median is the middle value in the list of numbers. To easily find the median, numbers have to be listed in numerical order from smallest to largest. The mode is the value that occurs most often. If no number in the list is repeated, then there is no mode in the list. A trimmed mean is similar to arithmetic mean (average), but it trims any lower and upper values, which is influenced by extremely high or low data values, to get a better fit for data sets.